

EXPERT 1:

**Validation format for data collection instruments
(Sample Form for CVI)**

Title of the Instrument for validation: “ Survey Instrument for Evaluating IoT-Enabled Smart Classroom Features in Universities”.

Dear Experts,

We beseech your expert opinion on how relevant each item is being measured. The instructions are provided to guide you in your analysis. Please use the following rating scale and rate each item as objective as possible in your review.

Instruction: This instrument used for the EXPERT EVALUATOR to assess the relevance and effectiveness of the instrument that is validating. You must place the punctuation that you consider appropriate to the different statements according to the following scale:

1:Not Relevant 2:Somewhat Relevant 3:Quite Relevant 4:High Relevant

Questionnaire Item (IoT-Enabled Smart Classroom Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed that IoT-enabled features such as wifi, interactive whiteboards, smart projectors, and online attendance systems are used in my class.				/	
I frequently use IoT-enabled smart classroom features, such as Wi-Fi, interactive whiteboards, smart projectors, and				/	

online attendance systems.					
I believe that smart classroom features are important in enhancing learning experiences.				/	

Questionnaire Item (Quality and Effectiveness of Smart Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I believe the availability of WiFi, interactive whiteboards, smart projectors, and online attendance systems creates flexible class environments.				/	
I have experienced technical issues with these features (eg. wi-fi slow, collaborative learning tools not functional).				/	
I feel that the features of a smart classroom adequately meet my learning needs.				/	
I find smart classroom features helpful in supporting collaboration, engagement,				/	

interactive learning, and communication.					
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Questionnaire Item (Impact on Academic Performance and Engagement)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed an improvement in academic performance since the implementation of smart classroom features, as they enhance engagement, collaboration, and overall learning efficiency.			/		Remove this “as they enhance engagement, collaboration, and overall learning efficiency”. You can create another statement for this.
I have found that smart classroom features have significantly enhanced my engagement through interactive and collaborative learning experiences.				/	
I am more motivated to participate in				/	

classes equipped with smart classroom features compared to traditional classrooms.					
I have faced challenges in using IoT-enabled tools in classrooms.				/	

Signature and name of Evaluator: **Amirul** Date: 20/12/24

EXPERT 2

Validation format for data collection instruments
(Sample Form for CVI)

Title of the Instrument for validation: “ Survey Instrument for Evaluating IoT-Enabled Smart Classroom Features in Universities”.

Dear Experts,

We beseech your expert opinion on how relevant each item is being measured. The instructions are provided to guide you in your analysis. Please use the following rating scale and rate each item as objective as possible in your review.

Instruction: This instrument used for the EXPERT EVALUATOR to assess the relevance and effectiveness of the instrument that is validating. You must place the punctuation that you consider appropriate to the different statements according to the following scale:

1:Not Relevant 2:Somewhat Relevant 3:Quite Relevant 4:High Relevant

Questionnaire Item (IoT-Enabled Smart Classroom Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed that IoT-enabled features such as wifi, interactive whiteboards, smart projectors, and online attendance systems are used in my class.				/	
I frequently use IoT-enabled smart classroom features, such as Wi-Fi, interactive whiteboards, smart projectors, and			/		

online attendance systems.					
I believe that smart classroom features are important in enhancing learning experiences.				/	

Questionnaire Item (Quality and Effectiveness of Smart Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I believe the availability of WiFi, interactive whiteboards, smart projectors, and online attendance systems creates flexible class environments.				/	
I have experienced technical issues with these features (eg. wi-fi slow, collaborative learning tools not functional).				/	
I feel that the features of a smart classroom adequately meet my learning needs.			/		
I find smart classroom features helpful in supporting collaboration, engagement,				/	

interactive learning, and communication.					
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Questionnaire Item (Impact on Academic Performance and Engagement)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed an improvement in academic performance since the implementation of smart classroom features, as they enhance engagement, collaboration, and overall learning efficiency.				/	
I have found that smart classroom features have significantly enhanced my engagement through interactive and collaborative learning experiences.				/	
I am more motivated to participate in classes equipped with smart classroom features compared to traditional classrooms.				/	

I have faced challenges in using IoT-enabled tools in classrooms.			/		
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Signature and name of Evaluator: Athirah Date: 17 December 2024

EXPERT 3

Validation format for data collection instruments (Sample Form for CVI)

Title of the Instrument for validation: "Survey Instrument for Evaluating IoT-Enabled Smart Classroom Features in Universities".

Dear Experts,

We beseech your expert opinion on how relevant each item is being measured. The instructions are provided to guide you in your analysis. Please use the following rating scale and rate each item as objective as possible in your review.

Instruction: This instrument used for the EXPERT EVALUATOR to assess the relevance and effectiveness of the instrument that is validating. You must place the punctuation that you consider appropriate to the different statements according to the following scale:

1:Not Relevant 2:Somewhat Relevant 3:Quite Relevant 4:High Relevant

Questionnaire Item (IoT-Enabled Smart Classroom Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed that IoT-enabled features such as wifi, interactive whiteboards, smart projectors, and online attendance systems are used in my class.				✓	
I frequently use IoT-enabled smart classroom features, such as Wi-Fi, interactive whiteboards, smart projectors, and				✓	

online attendance systems.					
I believe that smart classroom features are important in enhancing learning experiences.				✓	

Questionnaire Item (Quality and Effectiveness of Smart Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I believe the availability of WiFi, interactive whiteboards, smart projectors, and online attendance systems creates flexible class environments.				✓	
I have experienced technical issues with these features (eg. wi-fi slow, collaborative learning tools not functional).				✓	
I feel that the features of a smart classroom adequately meet my learning needs.				✓	
I find smart classroom features helpful in supporting collaboration, engagement,				✓	

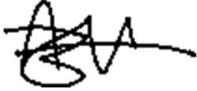
interactive learning, and communication.					
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Questionnaire Item (Impact on Academic Performance and Engagement)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed an improvement in academic performance since the implementation of smart classroom features, as they enhance engagement, collaboration, and overall learning efficiency.				✓	
I have found that smart classroom features have significantly enhanced my engagement through interactive and collaborative learning experiences.				✓	
I am more motivated to participate in classes equipped with smart classroom features compared to traditional classrooms.				✓	

I have faced challenges in using IoT-enabled tools in classrooms.				✓	
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Signature and name of Evaluator: _____

Date: 19/12/2024



(Nor Azwanadia binti Zolkepli)

EXPERT 4

Validation format for data collection instruments (Sample Form for CVI)

Title of the Instrument for validation: “ Survey Instrument for Evaluating IoT-Enabled Smart Classroom Features in Universities”.

Dear Experts,

We beseech your expert opinion on how relevant each item is being measured. The instructions are provided to guide you in your analysis. Please use the following rating scale and rate each item as objective as possible in your review.

Instruction: This instrument used for the EXPERT EVALUATOR to assess the relevance and effectiveness of the instrument that is validating. You must place the punctuation that you consider appropriate to the different statements according to the following scale:

1:Not Relevant 2:Somewhat Relevant 3:Quite Relevant 4:High Relevant

Questionnaire Item (IoT-Enabled Smart Classroom Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed that IoT-enabled features such as wifi, interactive whiteboards, smart projectors, and online attendance systems are used in my class.			/		
I frequently use IoT-enabled smart classroom features, such as Wi-Fi, interactive whiteboards, smart projectors, and			/		

online attendance systems.					
I believe that smart classroom features are important in enhancing learning experiences.				✓	

Questionnaire Item (Quality and Effectiveness of Smart Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I believe the availability of WiFi, interactive whiteboards, smart projectors, and online attendance systems creates flexible class environments.				✓	
I have experienced technical issues with these features (eg. wi-fi slow, collaborative learning tools not functional).			/		
I feel that the features of a smart classroom adequately meet my learning needs.				✓	
I find smart classroom features helpful in supporting collaboration, engagement,				✓	

interactive learning, and communication.					
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Questionnaire Item (Impact on Academic Performance and Engagement)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed an improvement in academic performance since the implementation of smart classroom features, as they enhance engagement, collaboration, and overall learning efficiency.			/		
I have found that smart classroom features have significantly enhanced my engagement through interactive and collaborative learning experiences.				✓	
I am more motivated to participate in classes equipped with smart classroom features compared to traditional classrooms.				✓	

I have faced challenges in using IoT-enabled tools in classrooms.			/		
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Signature and name of Evaluator: _____ Date:
 20/12/2024
 (RAJA ASNA SYAHINDAH)

EXPERT 5

Validation format for data collection instruments (Sample Form for CVI)

Title of the Instrument for validation: “ Survey Instrument for Evaluating IoT-Enabled Smart Classroom Features in Universities”.

Dear Experts,

We beseech your expert opinion on how relevant each item is being measured. The instructions are provided to guide you in your analysis. Please use the following rating scale and rate each item as objective as possible in your review.

Instruction: This instrument used for the EXPERT EVALUATOR to assess the relevance and effectiveness of the instrument that is validating. You must place the punctuation that you consider appropriate to the different statements according to the following scale:

1:Not Relevant 2:Somewhat Relevant 3:Quite Relevant 4:High Relevant

Questionnaire Item (IoT-Enabled Smart Classroom Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed that IoT-enabled features such as wifi, interactive whiteboards, smart projectors, and online attendance systems are used in my class.				✓	
I frequently use IoT-enabled smart classroom features, such as Wi-Fi, interactive whiteboards, smart projectors, and				✓	

online attendance systems.					
I believe that smart classroom features are important in enhancing learning experiences.				✓	

Questionnaire Item (Quality and Effectiveness of Smart Features)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I believe the availability of WiFi, interactive whiteboards, smart projectors, and online attendance systems creates flexible class environments.				✓	
I have experienced technical issues with these features (eg. wi-fi slow, collaborative learning tools not functional).				✓	
I feel that the features of a smart classroom adequately meet my learning needs.				✓	
I find smart classroom features helpful in supporting collaboration, engagement,				✓	

interactive learning, and communication.					
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Questionnaire Item (Impact on Academic Performance and Engagement)	Scale				Comment
	1 (Not Relevant)	2 (Somewhat Relevant)	3 (Quite Relevant)	4 (High Relevant)	
I have noticed an improvement in academic performance since the implementation of smart classroom features, as they enhance engagement, collaboration, and overall learning efficiency.			/		
I have found that smart classroom features have significantly enhanced my engagement through interactive and collaborative learning experiences.			✓		
I am more motivated to participate in classes equipped with smart classroom features compared to traditional classrooms.			✓		

I have faced challenges in using IoT-enabled tools in classrooms.		✓			
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20/12/2024
(SITI AISHAH)